

■ **Document Classification Alert:** This is an unedited AI conversation export, not a finished requirements document. It contains chatbot interface text, unfilled placeholder values, and at least one out-of-scope medical device specification. Do NOT commit to Doorstop without the editorial pass described in this stress test. Prepared for: **CEO · CTO** | Classification: Internal

Executive Summary

The URMOSES Extended Requirements document contains genuinely valuable engineering content — particularly the 62+ ergonomic specifications, the safety system requirements, and the ESG reporting architecture. These are concrete, testable, and strategically differentiated. However, the document has a fundamental problem that must be stated plainly: it is a raw export of a Gemini AI conversation, published as a project document without editorial review. It contains chatbot interface text referencing the Gemini UI, six or more unfilled "X" placeholder values in technical specifications, a medical device requirement (arrhythmia detection) that triggers FDA regulatory scope, and a Doorstop implementation description that is technically incorrect. The content is valuable raw material. The document is not ready for any internal or external use in its current form.

Element	Rating	Key Observation
62+ Ergonomic Specifications	✓ Strong	H-point, door width, sill height, slip resistance — concrete and testable.
Safety System Requirements	✓ Strong	AEB threshold, sensor array, SOS button specs are differentiated and prüfbar.
REQ/SPEC / Master/Subset Structure	✓ Correct	Conceptually right for Doorstop; content hierarchy is sound.
Management Summary (p.13-14)	✓ Useful	Clean strategic synthesis from product to platform. Good CEO/CTO briefing base.
Gemini UI Text in Document Body	✗ Critical	Chatbot conversation artifacts published as project documentation.
"X" Placeholder Values (6+)	✗ Critical	Unverifiable, untestable — not valid requirements for Doorstop.
Arrhythmia SPEC — Medical Device	✗ Critical	Triggers FDA / EU MDR regulatory scope. Out of LA28 budget and timeline.
Doorstop YAML Structure Described	✗ Technical	Sequential file = one requirement mapping is incorrect. Will break toolchain.
EU Bike Lane vs. LADOT Standards	■ Risk	EU micro-mobility width rules applied to LA operational context — unvalidated.
"Crucial Note" Repeated 4x	■ Signal	Indicates AI uncertainty about own output, not a stable system description.
Personal Name References in REQs	■ Minor	"Michelle's vision" / "as requested by CEO" not valid requirement rationale.

✓ What Works — Valuable Raw Material

✓ PASS

62+ Ergonomic Specifications: Concrete and Testable

The senior-specific requirements are the strongest engineering content in the document. H-point range 500-700mm, minimum door clear width 800mm, sill height max 400mm, slip-resistance coefficient 0.6 (dry) / 0.4 (wet), grab handle placement at A-pillar and roof rail — these are specific, measurable, and directly verifiable during prototype validation. Once "X" values are filled and Gemini artifacts removed, this block is ready for Doorstop commit.

✓ PASS

Safety System Requirements: Differentiated and Defensible

The 62+-calibrated safety specs are strategically significant: AEB activation at 20% earlier threshold than standard automotive presets for senior reaction times; sensor array detecting obstacles as small as 25mm (walking cane diameter) within 1500mm radius; SOS button minimum 30mm diameter with high-tactile feedback. These are not generic safety requirements — they are demographically specific and will differentiate URMOSES in any regulatory or insurance evaluation.

✓ PASS

Management Summary (Pages 13-14): Good CEO/CTO Briefing Base

The final Management Summary correctly synthesizes the strategic shift from product-centric (the vehicle) to platform-centric (Urban Mobility + Energy Autonomy + Demographic Care). The five-pillar structure — Chassis, Solar, Ergonomics, Digital/ESG, Lifecycle — is the right architecture for a board-level requirements overview. With Gemini artifacts removed and personal name references replaced with role titles, this section is usable as a standalone briefing document.

✗ Critical Issues — Do Not Commit to Doorstop

✗ CRITICAL

Gemini Conversation Artifacts Published as Project Documentation

The document contains raw chatbot interface text that was never removed before publication. These are not editorial comments — they are Gemini UI responses referencing its own memory system and user interface:

"I'll do my best to remember this for our future interactions. To make sure Gemini follows these instructions in the future, you can add them to 'Your instructions for Gemini' here."
"Would you like me to generate..." / "Should I now move to..." / "Next Step: Would you like me to..." [appears 6+ times]

When this document is shared internally with Michelle, a technical partner, or any stakeholder, the immediate conclusion is: URMOSES strategic requirements were generated by an unreviewed AI chatbot and published without editorial ownership. Fix: Full editorial pass — remove every AI conversational element. The requirement content must be owned and signed off by the CTO, not attributed to a generative AI session.

✗ CRITICAL

"X" Placeholder Values — Requirements Are Not Requirements

At least six technical specifications contain "X" as the value, rendering them unverifiable and uncommittable to Doorstop:

SPEC_001.1: "minimum energy yield of X kWh per day" REQ_001.3: "curb weight limited to X kg"
SPEC_001.2: "curvature radius must not exceed X mm" SPEC Subset: "maximum operating force of X Newtons" SPEC Subset: "minimum energy density of X Wh/kg" SPEC Subset: "transmit data at intervals of X minutes"

A Doorstop requirement with "X" as a value cannot be verified, cannot be tested, and cannot be certified. These are templates, not specifications. Each "X" must be replaced with a researched, CTO-validated value before commit. Sources: Cruise N Comfort USA datasheets (AC power), RIGID DV1910E-AC specs (energy density), solar manufacturer PV output data (kWh/day), LADOT MDS transmission requirements (interval).

✗ CRITICAL

Arrhythmia Detection SPEC — Medical Device Regulatory Scope

The document includes as a SPEC:

"The seat sensors must be capable of detecting irregular heart rate patterns (Arrhythmia) and triggering a Level_1 alert to the fleet operator."

Arrhythmia detection is a clinical diagnostic function regulated as a medical device under the US Federal Food, Drug and Cosmetic Act (FDA 510(k) clearance or PMA approval required) and the EU Medical Device Regulation (MDR 2017/745, Class IIb minimum). Implementing this as a vehicle SPEC triggers: — FDA pre-market submission process (12-24 months minimum) — ISO 13485 Quality Management System for medical devices — Clinical validation studies with patient data — CE marking as a medical device — Liability framework shift from mobility to healthcare None of this is in the \$8.7M budget or the LA28 timeline. Fix: Move to a clearly labeled "Post-2028 Platform Vision" backlog. Replace in LA28 scope with: "Seat sensors shall detect passenger presence and alert Mobility Buddy if no movement is detected for X minutes."

✗ TECHNICAL

Doorstop YAML File Structure Is Incorrectly Described

The document describes the Doorstop implementation as follows:

specifications/001.yml → REQ_001 specifications/002.yml → REQ_001.1 specifications/003.yml → REQ_001.2 specifications/004.yml → REQ_001.3 specifications/005.yml → SPEC_001

This is incorrect Doorstop architecture. In Doorstop, each .yml file contains exactly one requirement item identified by its ID field. Parent-child (hierarchical) relationships are defined via the "links:" field in the YAML, not by sequential file numbering. Sequential numbering of files is a valid storage convention, but the mapping "file 002 = child of file 001" does not exist in Doorstop — the toolchain resolves hierarchy through links, not filenames. If implemented as described, doorstop review and doorstop publish will not correctly resolve the REQ_001 / REQ_001.1 hierarchy. Fix: CTO to validate YAML structure against the established Doorstop workflow before first commit. The correct structure uses explicit links: [REQ_001] in each child item.

■ Risk Items — Resolve Before Requirements Baseline

■ RISK

EU Bike Lane Standards Applied to Los Angeles Operations

REQ_001 sets "100% compliance with EU micro-mobility and heavy-duty pedelec width/weight regulations" as the core competitive constraint, with 110cm maximum width derived from EU standards. URMOSSES operates primarily in Los Angeles, where bike lane standards are defined by LADOT — not EU directives. California Vehicle Code and LADOT design standards classify electric cargo cycles differently from EU heavy-duty pedelec categories. The 110cm width limit may be correct, conservative, or irrelevant for LADOT — but this has not been validated. If REQ_001 is committed to Doorstop with EU regulatory rationale for a LA28 deployment, it creates a compliance reference to inapplicable standards. Fix: CTO to confirm applicable LADOT vehicle classification and width constraints before REQ_001 is baselined. Add LADOT standard reference alongside or replacing EU reference.

■ SIGNAL

"Crucial Implementation Note" Repeated Four Times

The identical warning — "The identifiers used below are for visual organization only. The actual ID assignment... must be performed exclusively by the Doorstop logic to prevent system corruption" — appears on four separate pages of the document. This repetition is a signal, not a feature. It indicates that the AI generating the content was uncertain about its own output and inserted a protective disclaimer each time it produced requirement-like structures. For a reader unfamiliar with the generation context, four identical warnings suggest the system is fragile and the document author does not trust their own content. Fix: One clear implementation note at the document header. Remove the remaining three. The content should stand on its own authority once the CTO has reviewed and signed off.

■ MINOR

Personal Name References as Requirement Rationale

Requirements reference "Michelle's modular vision," "as requested by the CEO," and "Michelle's note on scaling, valuation, and long-term competitiveness." In a formal requirements management system like Doorstop, the rationale for a requirement must reference a stakeholder role, a standard, or a verifiable source — not a named individual's conversational note. This creates two problems: (1) the requirement rationale is not auditable if Michelle's role changes; (2) it exposes internal decision-making attribution if the document is ever shared externally. Fix: Replace "Michelle's vision" with "CEO Strategic Direction (Feb 2026)" or reference the specific business plan section. Keep people out of requirement text.

Editorial Action Plan — Before Doorstop Commit

#	Action	Owner	Priority
1	Remove all Gemini UI text, "Would you like me to..." prompts, and memory references.	CTO	IMMEDIATE
2	Move Arrhythmia SPEC to Post-2028 backlog. Replace with non-medical passive-legality detection.	CTO + Legal	IMMEDIATE
3	Fill all 6+ "X" placeholder values with researched, validated specifications.	CTO	BEFORE COMMIT
4	Validate Doorstop YAML structure: confirm links: field used for hierarchy, not title sequence.	CTO	BEFORE COMMIT
5	Confirm LADOT vehicle classification for REQ_001 width/weight constraints. Update standard reference.	CTO	BEFORE COMMIT
6	Reduce "Crucial Implementation Note" from 4 instances to 1 (document header only).	CTO	BEFORE COMMIT
7	Replace all personal name references in REQ/SPEC rationale with role titles and plan section refs.	CTO + Legal	BEFORE COMMIT
8	CTO sign-off on full requirement set as owner before any Doorstop baseline is created.	CTO	BEFORE COMMIT

What to Preserve — The Raw Material Worth Keeping

Content Block	Status	Next Step
62+ Ergonomic REQ/SPEC block	Keep — fill X values	Validate H-point, sill, door width against DIN/ISO ergonomic standards.
Safety System REQ/SPEC block	Keep — remove arrhythmia	Do not commit AEB, SOS, sensor array specs after CTO value validation.
ESG / Fleet Management REQ/SPEC block	Keep — fill X values	Define telematics interval, AES-256 confirmation, 48h buffer spec.
Manufacturing / LCA REQ block	Keep as-is	ISO 14040 reference is correct. 35% recycled material target is concrete.
Management Summary (p.13-14)	Keep — minor edit	Remove name references, use as CEO/CTO board briefing annex.

Overall Verdict: Valuable raw material — not a publishable document. The engineering content (ergonomics, safety, ESG, lifecycle) is substantive and worth preserving. The Arrhythmia SPEC must be removed from LA28 scope immediately — it is the single highest-risk item in the entire URMOSSES document suite from a regulatory standpoint. Do not commit anything to Doorstop until the CTO has completed the 8-item editorial pass above and signed off on the requirement set as its named owner.